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Olander

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(54) **PIPE LOOSENING DEVICE AND METHOD**

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(58) **Field of Classification Search**

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See application file for complete search history.

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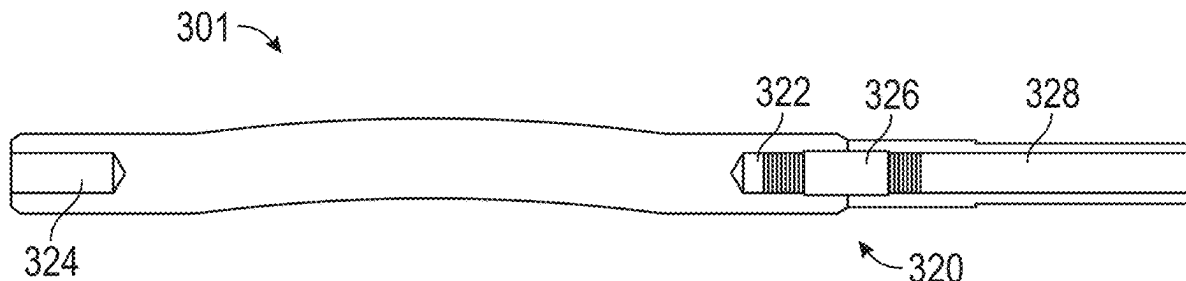
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(57) **ABSTRACT**

A pipe loosening device and methods are shown. In one example, the pipe loosening device enables a method wherein a pipe to be replaced is deformed but not burst, and a portion of a sidewall of the pipe is forced outward into the surrounding soil to compact a local region of the surrounding soil and loosen the existing pipe. The pipe may then be pulled from the ground along an axis of the pipe.

17 Claims, 5 Drawing Sheets



Related U.S. Application Data

Aug. 20, 2019, now Pat. No. 10,883,627, which is a division of application No. 15/865,031, filed on Jan. 8, 2018, now Pat. No. 10,422,441.

- (60) Provisional application No. 62/444,158, filed on Jan. 9, 2017.

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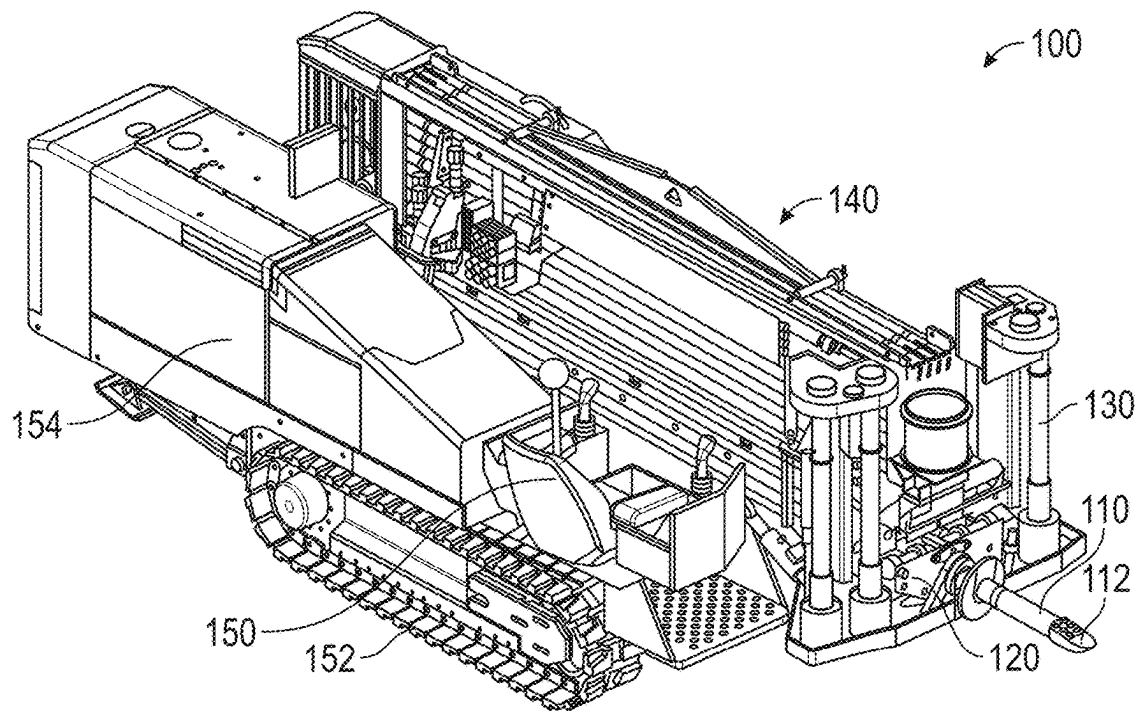


FIG. 1

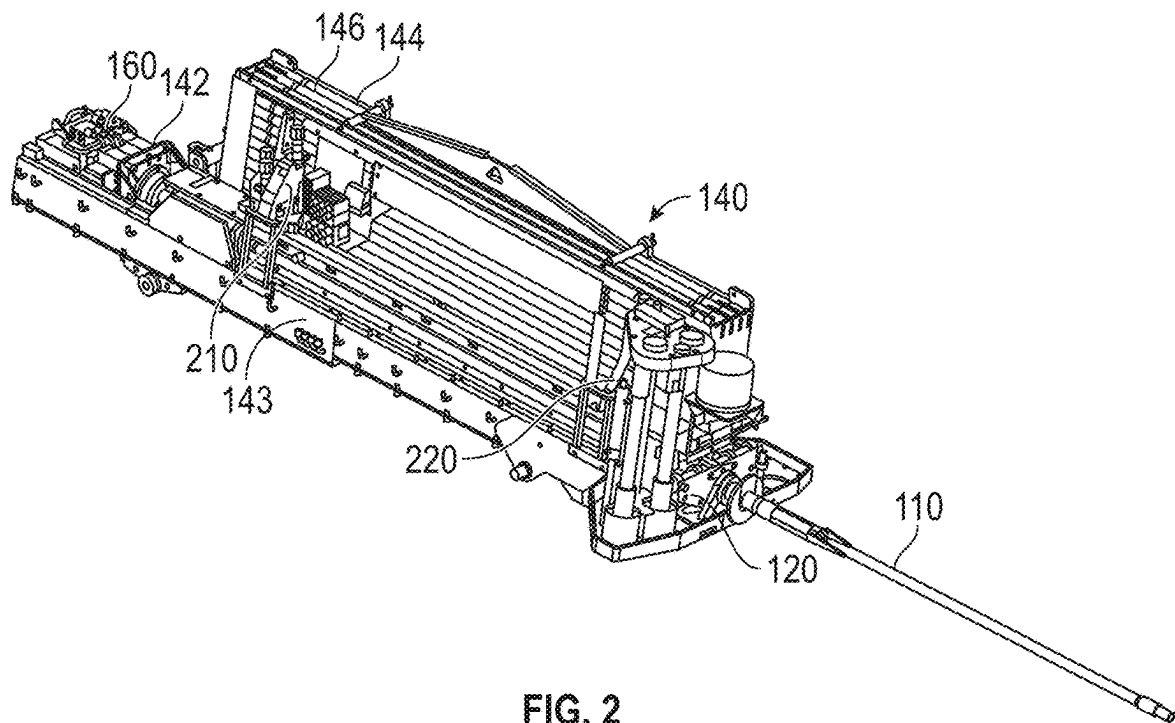


FIG. 2

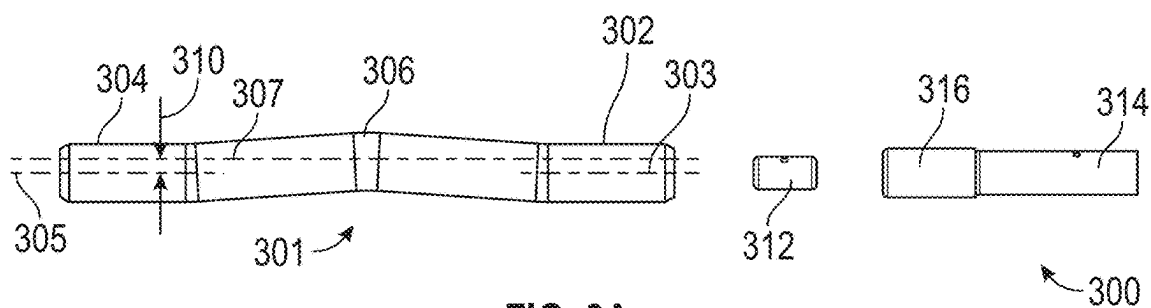


FIG. 3A

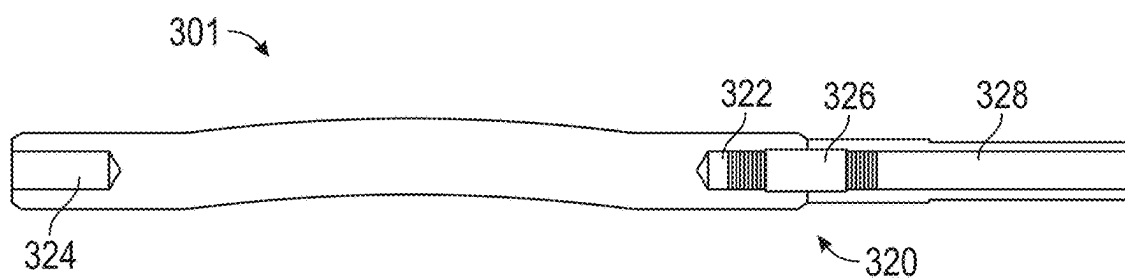


FIG. 3B

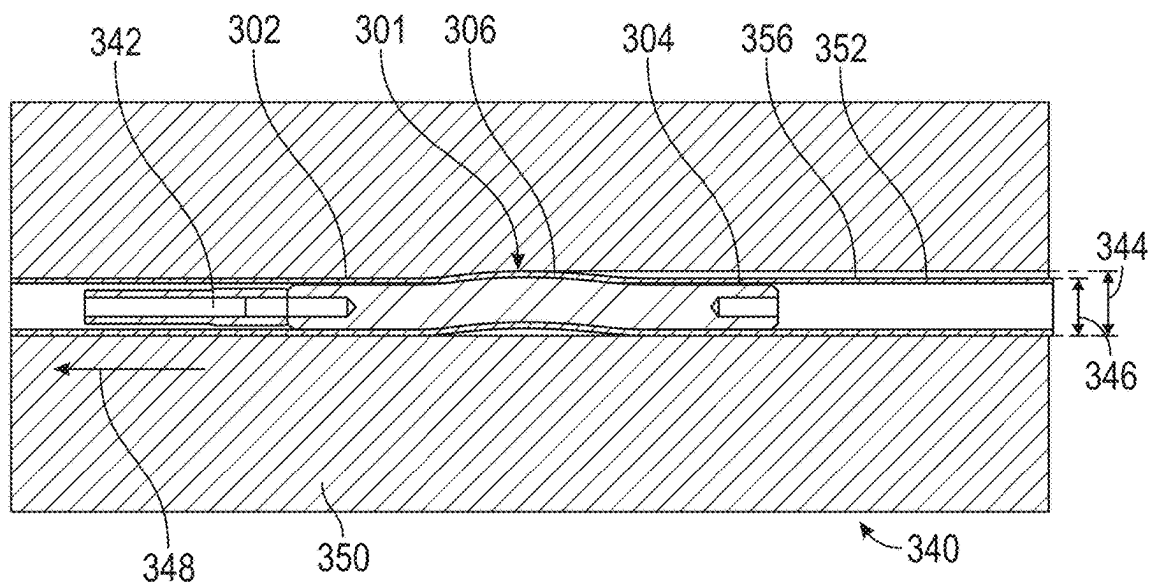


FIG. 3C

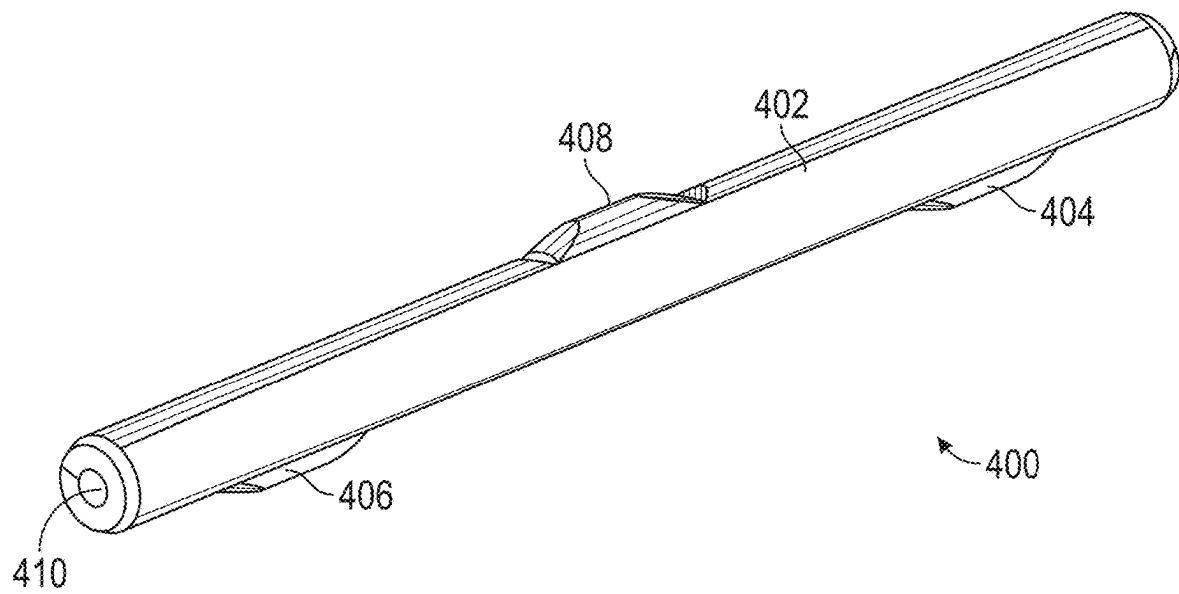


FIG. 4

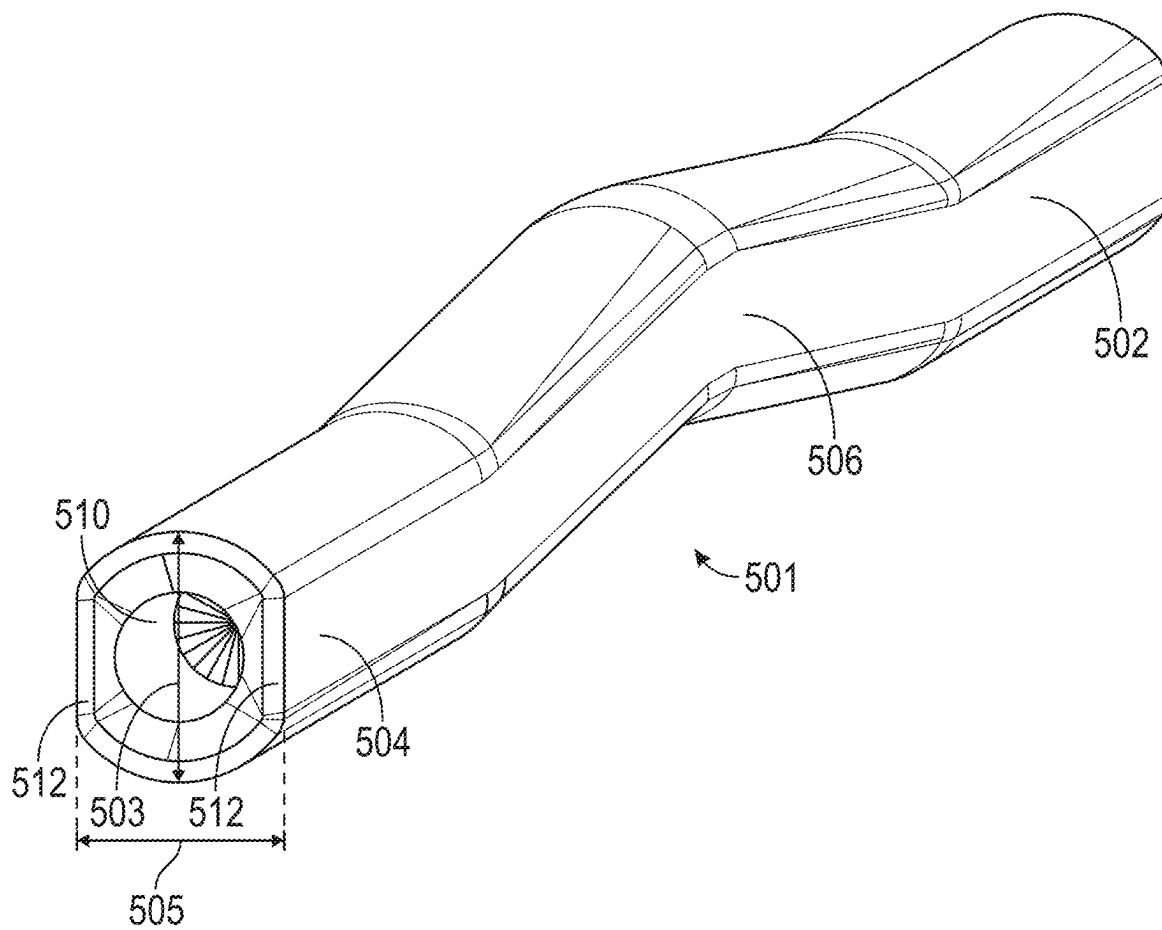


FIG. 5

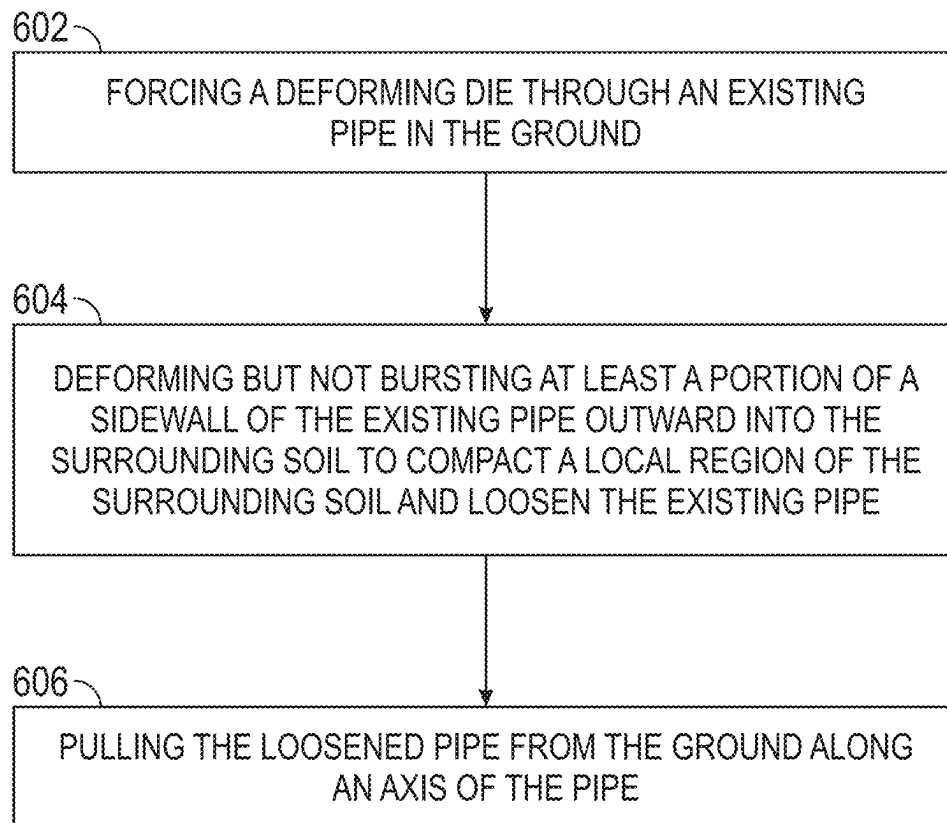


FIG. 6

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PIPE LOOSENING DEVICE AND METHOD

PRIORITY

This application is a continuation of U.S. patent application Ser. No. 17/697,468, filed Mar. 17, 2022, now issued as U.S. Pat. No. 11,982,394, which is a continuation of U.S. patent application Ser. No. 16/952,966, filed Nov. 19, 2020, now issued as U.S. Patent No. 11,293,579, which is a continuation of U.S. patent application Ser. No. 16/545,333, filed Aug. 20, 2019, now issued as U.S. Pat. No. 10,883,627 which is a division of U.S. patent application Ser. No. 15/865,031, filed Jan. 8, 2018, now issued as U.S. Pat. No. 10,422,441 which claims the benefit of priority to U.S. Provisional Application Ser. No. 62/444,158, filed Jan. 9, 2017, all of which are incorporated by reference herein in their entirety.

TECHNICAL FIELD

Embodiments described herein generally relate to pipe removal from underground. Specific examples may include using a die to loosen a pipe within soil before removing the pipe from the ground.

BACKGROUND

Pipe bursting is an existing technique where an old pipe is replaced by bursting the old pipe into the surrounding soil. A new pipe is then pulled into place where the old burst pipe once was. This technique is desirable because the old pipe may be replaced without digging a trench to access it. However, in some instances it may be desirable to replace an old pipe trenchlessly, but it may also be desirable to remove the old pipe from the ground. One example includes replacement of lead pipes. It is desirable to replace the pipe trenchlessly to avoid disturbing top surface structures such as roadways. However it is desirable to remove the lead pipe completely because lead may be hazardous to the environment.

Devices and methods are desired to accomplish these and other goals.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a directional drill in accordance with some embodiments of the invention.

FIG. 2 is a portion of a directional drill in accordance with some embodiments of the invention.

FIG. 3A is a pipe loosening device in accordance with some embodiments of the invention.

FIG. 3B is another pipe loosening device in accordance with some embodiments of the invention.

FIG. 3C is a pipe loosening device in use in accordance with some embodiments of the invention.

FIG. 4 is another pipe loosening device in accordance with some embodiments of the invention.

FIG. 5 is another pipe loosening device in accordance with some embodiments of the invention.

FIG. 6 is a flow diagram of a method of pipe extraction in accordance with some embodiments of the invention.

DESCRIPTION OF EMBODIMENTS

The following description and the drawings sufficiently illustrate specific embodiments to enable those skilled in the art to practice them. Other embodiments may incorporate

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structural, logical, electrical, process, and other changes. Portions and features of some embodiments may be included in, or substituted for, those of other embodiments. Embodiments set forth in the claims encompass all available equivalents of those claims.

A number of forcing equipment may be used with pipe loosening devices according to examples described in the present disclosure. In one example, a deforming die is pushed through a pipe to be replaced, as will be described in greater detail in examples below. One example of a pushing device includes a pneumatic hammer. Another example of a pushing device includes a static rod pusher.

In one example, a deforming die is pulled through a pipe to be replaced, as will be described in greater detail in examples below. One example of a pulling device includes a winch and cable. Another example of a pulling device includes a static rod puller.

In one example a horizontal directional drill (HDD) may be used to either pull or push a deforming die. One of ordinary skill in the art, having the benefit of the present disclosure, will recognize that the examples of forcing equipment (pulling and pushing) are not limiting. Other forcing equipment may also be used within the scope of the invention.

As discussed above, one example of forcing equipment includes a directional drill. FIG. 1 shows an example of a directional drill 100. The directional drill 100 includes a drill stem 110 including an attached sonde housing, and a drill head 112 for piercing the ground and leading a directional drill bore operation. A drill stem loader 140 is shown coupled to the directional drill 100. The drill stem loader 140 is configured to pick drill stem segments (or drill rods) from a drill stem magazine and add stem segments to the stem 110 during a boring operation. The drill stem loader 140 is further configured to remove stem segments from the drill stem 110 and replace them in the drill stem magazine after the boring operation is complete, and the drill stem is being retracted from the bore.

A power supply 154 is coupled to the directional drill 100 to drive the drill stem 110, and to operate other aspects of the directional drill 100. A cockpit 150 is further included in the directional drill 100, the cockpit 150 including a number of controllers and gauges to control and monitor a drilling operation. In one example, a track system 152 is included on the directional drill 100 to move and position the directional drill 100. A stake down system 130 is also shown coupled to a front end of the directional drill 100 in the example of FIG. 1. A drill stem vice 120 is further shown at a front end of the directional drill 100. Additional aspects of the drill stem vice 120 are described in more detail below.

FIG. 2 shows a portion of a directional drill 100 from FIG. 1, with a number of components removed to reveal more detail of a directional drill stem loader 140. The drill stem loader 140 includes a drill stem magazine 144, having a number of individual drill stem segments 146 loaded into the magazine 144.

A first linear actuator 210 and a second linear actuator 220 are shown adjacent to the drill stem magazine 144. In one example, the linear actuators 210, 220 are coupled to a pair of drill stem grippers as described in more detail below. Although two linear actuators are shown, the invention is not so limited. Other configurations may include a single linear actuator, or more than two linear actuators.

A drill head 142 is shown at a rear of the drill stem loader 140. The drill head 142 is mounted to a carriage frame 143 along a movable track. In one example, a drill fluid supply system 160 is coupled to the directional drill 100, adjacent

to the drill head **142**. During a drilling operation, the drill head **142** is operated to both rotate the drill stem **110**, and to drive the drill stem **110** forward into the ground. The drill stem vice **120** is shown at a front end of the drill stem loader **140**. During a drilling operation, the drill stem vice **120** selectively holds or releases individual segments of the drill stem **110** to aid in the adding or removal of drill stem segments (by screwing or unscrewing a threaded joint at either end of the drill stem segment).

FIG. **3A** shows a pipe loosening device **300** according to one example. The pipe loosening device **300** includes a solid single deforming die **301** and a connection to forcing equipment. In the example of FIG. **3A**, the forcing equipment is a cable **314** that is coupled to the deforming die **301** with a coupler **312**. In one example, the cable **314** is pulled by a winch (not shown). Other cable pulling equipment, such as a cyclic puller, may also be used.

The solid single deforming die **301** includes a leading end cylinder portion **302** and a trailing end cylinder portion **304**. The leading end cylinder portion **302** includes a leading end axis **303**, and the trailing end cylinder portion **304** includes a trailing end axis **305**. The leading end axis **303** and the trailing end axis **305** are substantially aligned.

In one example, both the leading end cylinder portion **302** and the trailing end cylinder portion **304** have a first diameter. In one example, the first diameter substantially matches an inner diameter of a pipe to be replaced. In one example, the leading end cylinder portion **302** and the trailing end cylinder portion **304** slide within the pipe to be replaced with a close tolerance fit. In one example the close tolerance fit serves to guide the solid single deforming die **301** and determines how a deforming force is applied using a deforming cylinder portion **306**.

As noted, the solid single deforming die **301** includes a deforming cylinder portion **306** coupled between the leading end cylinder portion **302** and the trailing end cylinder portion **304**. The deforming cylinder portion **306** includes an axis **307** that is displaced by a distance **310** away from the trailing end axis and the leading end axis. In one example, the deforming cylinder portion **306** is coupled between the leading end cylinder portion **302** and the trailing end cylinder portion **304** using a gentle transition or slope. In one example, this configuration helps to gradually apply a deforming force to a sidewall of the pipe to be replaced.

In one example, one or more of the leading end cylinder portion **302**, the trailing end cylinder portion **304**, and the deforming cylinder portion **306** includes a material that is hardened to resist galling or cold welding on the inside of the pipe to be replaced. In one example, one or more of the leading end cylinder portion **302**, the trailing end cylinder portion **304**, and the deforming cylinder portion **306** is formed from a hardened steel. In one example a hardness is within a range of 60-65 Rockwell C. In one example, one or more of the leading end cylinder portion **302**, the trailing end cylinder portion **304**, and the deforming cylinder portion **306** is case hardened, carburized, or otherwise coated to have a hard external surface.

In one example, a hardened insert is mechanically attached to wear points on one or more of the leading end cylinder portion **302**, the trailing end cylinder portion **304**, and the deforming cylinder portion **306**. An example of attachment includes, but is not limited to, screws, bolts, welding, adhesives, dovetail joints etc. Examples of hardened inserts includes, but is not limited to, tungsten carbide, or other carbides, hardened tool steel, coated tool steel, etc.

In one example a low friction insert may be used. An example of a low friction insert includes, but is not limited

to an aluminum-bronze alloy. In the example of aluminum-bronze, such an alloy provides a low friction in relation to several pipe material, specifically black iron. In one example, a lubricant may be used to further reduce friction and any resulting galling or cold welding.

FIG. **3B** shows another example of a pipe loosening device **320** according to one example. The pipe loosening device **320** includes a solid single deforming die **301** and a connection to forcing equipment. In the example of FIG. **3B**, the forcing equipment is a drill stem **328** from a directional drill, such as directional drill **100** from FIGS. **1** and **2**. The drill stem **328** is shown coupled to the solid single deforming die **301** with a coupler **326**.

In the example of FIG. **3B**, a threaded connection **322** is shown on the leading end cylinder portion **302**. In one example, a second threaded connection **324** is shown on the trailing end cylinder portion **304**. Various combinations of threaded connections may be used in different combinations. In the example shown in FIG. **3B**, the solid single deforming die **301** is flexible in the method of connection to forcing equipment. Either pushing or pulling equipment may be used. In the example, shown, the threaded connections **322**, **324** are both female threads. However, the invention is not so limited. One or both of the threaded connections **322**, **324** may be male threads instead. In one example, one or more of the threaded connections **322**, **324** may include a tapered thread, to mate with a tapered thread on a directional drill stem.

FIG. **3C** shows another example of a pipe loosening device **340** according to one example. The pipe loosening device **320** includes a solid single deforming die **301** and a connection to forcing equipment. In the example of FIG. **3C**, the forcing equipment is a drill stem **342** from a directional drill, such as directional drill **100** from FIGS. **1** and **2**.

The pipe loosening device **340** in FIG. **3C** is shown in operation within a pipe to be replaced **352**. The pipe to be replaced is shown embedded within soil **350**. In operation, the solid single deforming die **301** is pulled or pushed in direction **348**. The diameter of the leading end cylinder portion **302** and the trailing end cylinder portion **304** substantially matches an inner diameter **346** of the pipe to be replaced **352**. Because of the substantial matching, the leading end cylinder portion **302** and the trailing end cylinder portion **304** guide the solid single deforming die **301** through the pipe to be replaced **352** in a controlled manner.

As the solid single deforming die **301** is pulled, the deforming cylinder portion **306** is forced against a sidewall of the pipe to be replaced **352**. The deforming cylinder portion **306** pushes the sidewall portion outward into the surrounding soil and causes a gap **356** to form between the sidewall of the pipe to be replaced **352** and the soil **350**. In one example, the pipe to be replaced **352** is not burst or in any other way split or opened to the soil **350**. Bursting, splitting, or otherwise opening up the pipe to be replaced **352** to the surrounding soil may cause additional unwanted friction when the pipe to be replaced **352** is later removed from the soil along an axis of the pipe. Bursting, splitting, or otherwise opening up the pipe to be replaced **352** may also weaken the pipe to be replaced, and cause it to break during the later extraction operation where the pipe to be replaced **352** is removed from the soil along an axis of the pipe.

FIG. **3C** shows how the gap **356** increases dimension **344** to an dimension that is larger than the diameter **346** of the pipe to be replaced. Although the invention is not so limited, in one example, the diameter of the pipe to be replaced is approximately 1.66 inches, and the dimension **344** after passing of the deforming cylinder portion **306** is approxi-

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mately 1.901 inches. In one specific example, for a $\frac{5}{8}$ inch copper OD pipe, the dimension **344** is about 13% of the OD. In another specific example, for a $1\frac{1}{2}$ inch steel OD pipe, the dimension **344** is about 11% of the OD. In another specific example, for a 2 inch steel OD pipe, the dimension **344** is about 9% of the OD.

In one example, in order to deform, but not burst the pipe to be replaced **352**, the deforming cylinder portion **306** includes an axis **307** that is displaced by a distance **310** away from the trailing end axis and the leading end axis that is between 5 and 15 percent of the diameter of the leading end cylinder portion **302** and the trailing end cylinder portion **304**. In one example, the displacement **310** is approximately 7 percent of the diameter of the leading end cylinder portion **302** and the trailing end cylinder portion **304**. In one example, the displacement **310** decreases as the OD of the pipe to be replaced increases.

As noted above, the diameter of the leading end cylinder portion **302** and the trailing end cylinder portion **304** will vary depending on the inner diameter of the pipe to be replaced **352**. However, in one example, the percent displacement of the distance **310** away from the trailing end axis and the leading end axis is proportional, depending on the diameter of the pipe to be replaced.

In one example, the passing of the solid single deforming die **301** through the pipe to be replaced **352** will loosen the pipe from within the soil **350**. after loosening, the pipe to be replaced **352** may be pulled from the ground along an axis of the pipe, such as in direction **348**. If the pipe is not loosened from the soil **350**, the pipe may tear, with portions of the pipe undesirably remaining in the soil **350**.

Any number of possible pipe types may be replaced using the devices and methods shown. Examples include, but are not limited to, lead pipes, copper pipes, natural gas pipes, etc. In one particular example, small diameter black iron gas pipe may be replaced using the devices and methods described above. In may be necessary to remove small diameter black iron gas pipe from the ground, instead of bursting the pipe, due to concerns for new plastic pipe being drawn in to the ground to replace the black iron pipe. In some instances, bursting may be inadequate to move the old pipe out of the way of the new pipe being drawn in. The old pipe may cut or damage the new pipe. In such an instance, it may be desirable to remove the old pipe from the ground.

FIG. 4 shows another pipe loosening device according to one example. A deforming die **401** is shown that may be used in a manner similar to the examples shown in FIGS. 3A-3C. Any suitable forcing equipment may be used, as described above, including but not limited to, cables, winches, cyclic pullers, pipe pushers, directional drills and drill stem, etc.

The deforming die **401** includes a cylinder portion **402**. In one example, the cylinder portion **402** have a first diameter. In one example, the first diameter substantially matches an inner diameter of a pipe to be replaced. In one example, the cylinder portion **402** slides within the pipe to be replaced with a close tolerance fit. In one example the close tolerance fit serves to guide the deforming die **401** and determines how a deforming force is applied using deforming inserts as described below.

FIG. 4 shows a first end insert **404**, a second end insert **406**, and a middle insert **408**. In one example, the inserts **404**, **406**, **408** include tungsten carbide inserts. Although tungsten carbide is used as an example, other hard materials may also be used. Examples include, but are not limited to, other carbide materials, hardened tool steels, ceramic materials, etc. Inserts may be secured to the cylinder portion **402**

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in a number of possible configurations, such as welding, or adhesive such as epoxy to secure inserts within a pocket. Inserts may also be secured using threaded fasteners through a hole from a side of the cylinder portion **402** opposite the inserts. In one example, the inserts are replaceable.

In use, when the protruding inserts force themselves against a sidewall of a pipe to be replaced, deformation occurs in the pipe to be replaced. The deformation is small enough to loosen soil that surrounds the pipe to be replaced, but the deformation is not large enough to tear the pipe to be replaced apart. In this way, the pipe to be replaced maintains enough integrity to be pulled from the ground without fracturing, but the loosened soil reduces friction on an outer surface of the pipe to be replaced, facilitating easier removal.

FIG. 4 further shows a central opening **410** that may be threaded or otherwise configured for attachment to forcing equipment such as a directional drill etc. as described above. Although three inserts **404**, **406**, **408** are shown, the invention is not so limited. Other numbers of inserts and other placement of inserts may also be used within the scope of the invention. Primarily, the inserts may be placed and configured to provide enough deformation to loosen a pipe to be replaced within surrounding soil, but the inserts may not provide so much deformation that the pipe to be replaced is significantly damaged.

FIG. 5 shows a deforming die **501** that may be used in a manner similar to the examples shown in FIGS. 3A-3C. Any suitable forcing equipment may be used, as described above, including but not limited to, cables, winches, cyclic pullers, pipe pushers, directional drills and drill stem, etc.

The deforming die **501** includes a leading end cylinder portion **502** and a trailing end cylinder portion **504**. The leading end cylinder portion **502** includes a leading end axis, and the trailing end cylinder portion **504** includes a trailing end axis. In the example of FIG. 5, the leading end axis and the trailing end axis are substantially aligned.

In one example, both the leading end cylinder portion **502** and the trailing end cylinder portion **504** have a first diameter **503**. In one example, the first diameter **503** substantially matches an inner diameter of a pipe to be replaced. In one example, the leading end cylinder portion **502** and the trailing end cylinder portion **504** slide within the pipe to be replaced with a close tolerance fit. In one example the close tolerance fit serves to guide the deforming die **501** and determines how a deforming force is applied using a deforming cylinder portion **506**.

In one example, the first diameter **503** is between about 90% and about 60% of an inner diameter of a pipe to be replaced. In one example, for inner diameters of a $1\frac{1}{4}$ to $1\frac{1}{2}$ inch pipe to be replaced, the first diameter **503** is about 85% of the inner diameter of the pipe to be replaced. In one example, for inner diameters of a 2 inch pipe to be replaced, the first diameter **503** is about 65% of the inner diameter of the pipe to be replaced.

The deforming die **501** includes a deforming cylinder portion **506** coupled between the leading end cylinder portion **502** and the trailing end cylinder portion **504**. The deforming cylinder portion **506** includes an axis that is displaced by a distance away from the trailing end axis and the leading end axis, similar to the example shown in FIGS. 3A-3C. In one example, the deforming cylinder portion **506** is coupled between the leading end cylinder portion **502** and the trailing end cylinder portion **504** using a gentle transition or slope. In one example, this configuration helps to gradually apply a deforming force to a sidewall of the pipe to be replaced.

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Similar to examples above, in order to deform, but not burst the pipe to be replaced, the deforming cylinder portion **506** includes an axis that is displaced by a distance away from the trailing end axis and the leading end axis that is between 5 and 15 percent of the diameter of the leading end cylinder portion **502** and the trailing end cylinder portion **504**. In one example, the displacement decreases as the OD of the pipe to be replaced increases.

FIG. **5** further shows a central opening **510** that may be threaded or otherwise configured for attachment to forcing equipment such as a directional drill, cable pulling, etc. as described above.

FIG. **5** further shows sides **512** perpendicular to the direction of deforming cylinder portion **506** offset. In the example shown, the sides **512** are reduced from the first diameter **503** to reduced width **505**. In one example, the reduced width **505** further facilitates deformation of the pipe to be replaced, allowing it to become more “oval” around the deforming die **501**. The oval cross section further enhances loosening of the pipe to be replaced within the soil. Although illustrated in FIG. **5**, the reduced width **505** configuration may be applied to any of the examples described in the present disclosure.

FIG. **6** shows a flow diagram of a method of pipe extraction according to an embodiment of the invention. In operation **602** a deforming die, such as the solid single deforming die **301**, deforming die **401**, or deforming die **501** is forced through an existing pipe in the ground. In operation **604**, at least a portion of a sidewall of the existing pipe is deformed but not burst outward into the surrounding soil to compact a local region of the surrounding soil and loosen the existing pipe. In operation **606**, the loosened pipe is pulled from the ground along an axis of the pipe.

To better illustrate the method and apparatuses disclosed herein, a non-limiting list of examples is provided here:

Example 1 includes a pipe loosening device. The pipe loosening device includes a solid single deforming die, including a leading end cylinder portion, having a first diameter, a trailing end cylinder portion, having a diameter substantially matching the first diameter, wherein a trailing end axis is substantially aligned with a leading end axis, and a deforming cylinder portion located between the leading end cylinder portion and the trailing end cylinder portion, wherein an axis of the deforming cylinder portion is parallel to, and displaced by a distance away from the trailing end axis and the leading end axis.

Example 2 includes the pipe loosening device of example 1, wherein the distance of displacement of the axis of the deforming cylinder portion from the trailing end axis and the leading end axis is between 5 and 10 percent of the first diameter.

Example 3 includes the pipe loosening device of any one of examples 1-2, wherein the distance of displacement of the axis of the deforming cylinder portion from the trailing end axis and the leading end axis is approximately 7 percent of the first diameter.

Example 4 includes the pipe loosening device of any one of examples 1-3, further including a threaded connection on the leading end cylinder portion that is adapted to couple to a directional drill rod section.

Example 5 includes the pipe loosening device of any one of examples 1-4, wherein the threaded connection is a female threaded connection.

Example 6 includes the pipe loosening device of any one of examples 1-5, wherein the threaded connection is a male threaded connection.

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Example 7 includes the pipe loosening device of any one of examples 1-6, further including a threaded connection on the trailing end cylinder portion that is adapted to couple to a directional drill rod section.

Example 8 includes the pipe loosening device of any one of examples 1-7, wherein the threaded connection is a female threaded connection.

Example 9 includes the pipe loosening device of any one of examples 1-8, wherein the threaded connection is a male threaded connection.

Example 10 includes a method of pipe extraction. The method includes forcing a deforming die through an existing pipe in the ground, deforming but not bursting at least a portion of a sidewall of the existing pipe outward into the surrounding soil to compact a local region of the surrounding soil and loosen the existing pipe, and pulling the loosened pipe from the ground.

Example 11 includes the method of example 10, wherein forcing the deforming die includes pulling with a directional drill stem.

Example 12 includes the method of any one of examples 10-11, wherein forcing the deforming die includes pulling with a winch.

Example 13 includes the method of any one of examples 10-12, wherein forcing the deforming die includes pushing with a directional drill stem.

Example 14 includes the method of any one of examples 10-13, wherein forcing the deforming die includes pushing with a rod pusher.

Example 15 includes the method of any one of examples 10-14, wherein forcing the deforming die includes pushing with a pneumatic hammer.

Example 16 includes the method of any one of examples 10-15, wherein deforming but not bursting at least a portion of a sidewall of the existing pipe includes deforming with a solid single deforming die, including a leading end cylinder portion, having a first diameter, a trailing end cylinder portion, having a diameter substantially matching the first diameter, wherein a trailing end axis is substantially aligned with a leading end axis, and a deforming cylinder portion located between the leading end cylinder portion and the trailing end cylinder portion, wherein an axis of the deforming cylinder portion is parallel to, and displaced by a distance away from the trailing end axis and the leading end axis.

Example 17 includes a pipe loosening device, including a deforming die, including a cylinder portion sized with a cylinder diameter to fit closely within an inner diameter of a pipe to be replaced, and one or more inserts coupled to a side of the cylinder portion, wherein the one or more inserts protrude from the cylinder portion by a distance.

Example 18 includes the pipe loosening device of example 17, wherein the distance is between 5 and 10 percent of the cylinder diameter.

Example 19 includes the pipe loosening device of any one of examples 17-18, wherein the distance is approximately 7 percent of the cylinder diameter.

Example 20 includes the pipe loosening device of any one of examples 17-19, further including a threaded connection on the cylinder portion that is adapted to couple to a directional drill rod section.

Example 21 includes the pipe loosening device of any one of examples 17-20, wherein the one or more inserts are formed from tungsten carbide.

Example 22 includes the pipe loosening device of any one of examples 17-21, wherein the one or more inserts include

two end inserts and a central insert on a side of the cylinder portion opposite the end inserts.

The above detailed description includes references to the accompanying drawings, which form a part of the detailed description. The drawings show, by way of illustration, specific embodiments in which the invention can be practiced. These embodiments are also referred to herein as “examples.” Such examples can include elements in addition to those shown or described. However, the present inventors also contemplate examples in which only those elements shown or described are provided. Moreover, the present inventors also contemplate examples using any combination or permutation of those elements shown or described (or one or more aspects thereof), either with respect to a particular example (or one or more aspects thereof), or with respect to other examples (or one or more aspects thereof) shown or described herein.

In this document, the terms “a” or “an” are used, as is common in patent documents, to include one or more than one, independent of any other instances or usages of “at least one” or “one or more.” In this document, the term “or” is used to refer to a nonexclusive or, such that “A or B” includes “A but not B,” “B but not A,” and “A and B,” unless otherwise indicated. In this document, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Also, in the following claims, the terms “including” and “comprising” are open-ended, that is, a system, device, article, composition, formulation, or process that includes elements in addition to those listed after such a term in a claim are still deemed to fall within the scope of that claim. Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects.

The above description is intended to be illustrative, and not restrictive. For example, the above-described examples (or one or more aspects thereof) may be used in combination with each other. Other embodiments can be used, such as by one of ordinary skill in the art upon reviewing the above description. The Abstract is provided to comply with 37 C.F.R. § 1.72(b), to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. Also, in the above Detailed Description, various features may be grouped together to streamline the disclosure. This should not be interpreted as intending that an unclaimed disclosed feature is essential to any claim. Rather, inventive subject matter may lie in less than all features of a particular disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate embodiment, and it is contemplated that such embodiments can be combined with each other in various combinations or permutations. The scope of the invention should be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

The invention claimed is:

1. A pipe loosening device, comprising:

a rigid deforming die, including:

a leading end body portion, having a first lateral dimension sized to fit closely within an inner diameter of a pipe to be replaced;

a trailing end body portion, having a second lateral dimension substantially matching the first lateral dimension, wherein a trailing end axis is substantially aligned with a leading end axis;

a protrusion located between the leading end body portion and the trailing end body portion, wherein the protrusion extends from the leading end body portion and the trailing end body portion by a distance between 5 and 10 percent of the first lateral dimension; and

a threaded connection on the leading end body portion that is adapted to couple to a directional drill rod section.

2. The pipe loosening device of claim 1, wherein the protrusion extends from the leading end body portion and the trailing end body portion by a distance of 7 percent of the first lateral dimension.

3. The pipe loosening device of claim 1, wherein the protrusion includes tungsten carbide.

4. The pipe loosening device of claim 1, further including two additional protrusions on a same side of the rigid deforming die, the two additional protrusions spaced apart from one another along the leading end axis, and wherein the protrusion extends on a side of the rigid deforming die substantially opposite the two additional protrusions, and wherein the protrusion is located longitudinally between the two additional protrusions.

5. The pipe loosening device of claim 4, wherein the two additional protrusions include tungsten carbide.

6. The pipe loosening device of claim 1, wherein the rigid deforming die includes a reduced width that is smaller than the first lateral dimension.

7. The pipe loosening device of claim 1, wherein the leading end body portion is cylindrical, and the trailing end body portion is cylindrical.

8. The pipe loosening device of claim 7, wherein the leading end body portion and the trailing end body portion both include a cylindrical cross section with a reduced width.

9. The pipe loosening device of claim 1, wherein the protrusion is included as part of a middle portion having a third lateral dimension substantially matching the first lateral dimension and second lateral dimension, wherein a middle axis of the middle portion is offset from the leading end axis and the trailing end axis by the distance.

10. The pipe loosening device of claim 1, further including a second threaded connection on a trailing end of the rigid deforming die.

11. A pipe loosening system, comprising:

a directional drill;

one or more drill stem sections coupled to the directional drill;

a rigid deforming die coupled to a distal end of the one or more drill stem sections, including:

a leading end body portion, having a first lateral dimension sized to fit closely within an inner diameter of a pipe to be replaced;

a trailing end body portion, having a second lateral dimension substantially matching the first lateral dimension, wherein a trailing end axis is substantially aligned with a leading end axis;

a protrusion located between the leading end body portion and the trailing end body portion, wherein the protrusion extends from the leading end body portion and the trailing end body portion by a distance between 5 and 10 percent of the first lateral dimension; and

a threaded connection on the leading end body portion that is adapted to couple to a directional drill rod section.

12. The pipe loosening system of claim 11, wherein the directional drill includes a drill stem loader.

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13. The pipe loosening system of claim **11**, wherein the directional drill includes a track system.

14. The pipe loosening system of claim **11**, wherein the protrusion includes tungsten carbide.

15. The pipe loosening system of claim **14**, wherein the rigid deforming die includes a reduced width that is smaller than the first lateral dimension. 5

16. The pipe loosening system of claim **11**, wherein leading end body portion is cylindrical, and the trailing end body portion is cylindrical. 10

17. The pipe loosening system of claim **16**, wherein the leading end body portion and the trailing end body portion both include a cylindrical cross section with a reduced width.

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